
Working Memory in the Chain of Thought Is Not Captured by J-Space

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 A language model’s internal workspace holds at most a few steps of chained rea-
2 soning state, at every scale we test, and the chain of thought holds the rest. A
3 common reading of J-space, the concept workspace a fitted Jacobian lens reads
4 from the residual stream, is that reasoning is carried internally and writing re-
5 lieves the workspace by externalizing state it would otherwise hold. Three mea-
6 surements contradict the relief picture. Instructed to answer chained arithmetic
7 without writing, models from 1.5B to 671B parameters collapse beyond one to two
8 dependent steps; a 671B model, so instructed, fails two chained additions. Models
9 write every intermediate value from the first step at every difficulty, so externaliza-
10 tion begins at no capacity threshold. The allocation between the two sides never
11 moves: damaging the workspace does not increase writing, capping writing does
12 not move computation inward, and ablating the model’s active J-space concepts
13 at the strongest dose that preserves ordinary generation changes nothing, even at
14 the one-to-two-step depths where the workspace does perform the arithmetic: the
15 memory the computation uses is not captured by that readout. We then show how
16 this memory works, with causal interventions on six unmodified models (used as
17 released, with no fine-tuning or task-specific training) from four families on arith-
18 metic and narrative state-tracking tasks. Each written value is held at its own token
19 position as a register: overwriting the hidden state there with the state a different
20 number would have produced, text unchanged, redirects the final answer to that
21 never-written number in 0.70 to 1.00 of cases, while size-matched random noise
22 redirects at most 0.01. The register is read through attention to its position (block-
23 ing it collapses dependence to at most 0.007 with matched controls unchanged),
24 the read completes by two thirds of the network’s depth, and two positions act as
25 independent stores whose planted values compose into a sum appearing nowhere
26 in the input (0.88 to 0.90). For oversight, identical visible text can carry different
27 stored values with different downstream behavior: a chain of thought exposes the
28 symbols of a computation, without guaranteeing their hidden content.

29 1 Introduction

30 The chain of thought is the primary working memory of serial reasoning in language models; the
31 internal workspace behind it holds at most a few dependent steps, and writing is not overflow from
32 it. We establish this claim with causal interventions on six unmodified models from four families
33 (Qwen, Phi, OLMo, DeepSeek-R1-Distill), on two task families with exact ground truth for every
34 intermediate value.

35 The capacity half addresses the overflow picture of externalization: reasoning state is carried inter-
36 nally, and writing takes over when the workspace can no longer hold it. This was our pre-registered

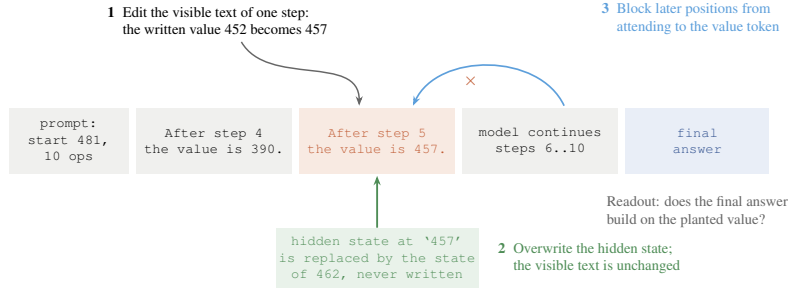


Figure 1: The edit-and-continue protocol. One written step of a correct worked solution is altered and the model continues from it; the readout is whether the final answer builds on the altered value. The interventions: (1) edit the visible text of the step; (2) overwrite the hidden state at the value’s token, leaving the text unchanged; (3) block attention from later positions to the value’s token.

hypothesis, and it is a natural reading of workspace accounts of reasoning, including J-space, the concept workspace a fitted Jacobian lens reads from the residual stream [Gurnee et al., 2026]. A workspace that writing relieves should show up in three places, and it shows up in none of them.

1. **No fallback.** Instructed to answer chained arithmetic with no writing (token counts confirm the channel is closed), models collapse beyond one to two dependent steps at every scale to 671B (Section 3).
2. **No threshold.** Models write 0.93 to 1.00 of ground-truth intermediates from one-step problems onward, at every scale from 1.5B to 671B (Section 3).
3. **No reallocation.** Damaging the workspace does not increase writing, capping writing does not move computation inward, and ablating the top active J-space concepts at a generation-preserving dose changes neither accuracy nor what gets written (Section 3).

The second half is the mechanism of the memory that remains.

4. **Registers.** The hidden state at a written value’s position is a settable value register: planting a never-written value there, text unchanged, redirects the answer at 0.70 to 1.00 across all six models (Section 4).
5. **Retrieval.** The register is read through attention to its position: blocking that attention collapses dependence to at most 0.007 on four models in three families, with matched control blocks unchanged; layer sweeps time the read to the middle third of the network (Section 5).
6. **Composition, and when the register is trusted.** Two positions are independently addressable, composing planted values into their sum (0.88 to 0.90). Which prompt evidence triggers recomputation is model-specific, and reliance on the trace tracks whether the model can regenerate the state, reconciling conflicting reports on how trace reliance scales (Sections 4 and 6).

Prior work anticipated the memory role of written tokens theoretically [Merrill and Sabharwal, 2024], behaviorally [Lanchantin et al., 2023], and causally on a fine-tuned model [Shih et al., 2026] and on multiplication and dynamic-programming tasks [Zhu et al., 2025], without testing whether an internal alternative exists, how the value is retrieved, or whether stores compose.

2 Experimental setup

2.1 Tasks and models

Both task families give exact ground truth for every intermediate value, so the downstream effect of any edit is computable. Arithmetic chains start from a three-digit number and apply d signed two-digit additions and subtractions. Narrative state-tracking problems describe a character whose count

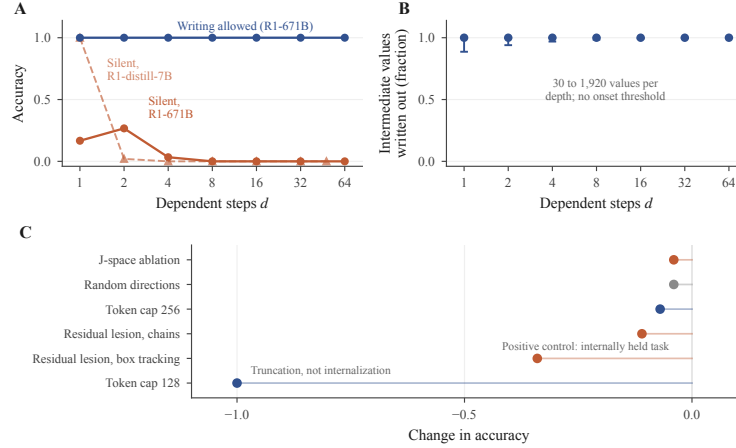


Figure 2: Writing is not triggered by exhausted internal capacity. (A) Instructed to answer without writing, models collapse after one to two dependent steps; with writing allowed, accuracy stays near ceiling ($n=30$ items per depth). (B) The fraction of intermediate values written is 1.00 at every depth (Wilson intervals), so writing begins at no threshold. (C) No intervention moves state between the workspace and the trace: J-space ablation and random directions leave accuracy unchanged; residual lesions hurt the internally held box-tracking task three times more than the written chain task; a token cap below the per-step floor collapses accuracy by truncation.

of an item changes through varied events, with varied names, items, verbs, and sentence forms. In both, the model writes one line per step giving the running value. White-box experiments use Qwen3-4B, Qwen2.5-7B-Instruct, Phi-3-medium, OLMo-2-7B-Instruct, and DeepSeek-R1-Distill-Qwen 7B and 14B; frontier models (DeepSeek V3.2, Claude Sonnet 4.5, Llama 3.3 70B) are tested behaviorally through assistant prefill.

2.2 Edit-and-continue protocol

The protocol takes a correct worked solution, changes the value at one step j , truncates immediately after the edited line, and lets the model continue (Figure 1). A no-edit truncation measures the sampling noise floor, 0.00 throughout. Generation uses temperature 0.6 and top- p 0.95 with five samples per item; unparseable outputs are counted separately. Intervals are 95 percent bootstrap intervals over items. Prompts and configurations are in the appendix; code and data will be released.

3 Does writing replace internal working memory?

3.1 Performance without written intermediates

Instructed to output only the final answer (Figure 2A), with token counts confirming nothing else was generated, models fail beyond one to two dependent steps (Qwen2.5-7B: 1.00 at $d=1$, 0.04 at $d=4$; DeepSeek V3.2 at 671B: 0.23 at $d=1$, near zero above), and the values used are random, so nothing can be recalled rather than computed.

3.2 When models write intermediate values

Models write 0.93 to 1.00 of ground-truth intermediates in free generation from $d=1$ through $d=64$, at every scale from 1.5B to 671B, so writing does not begin at any capacity threshold.

3.3 Tests for reallocation between internal and written state

A residual-stream lesion strong enough to cross the accuracy cliff makes traces disintegrate rather than expand; hard token budgets compress wording up to 2.5 times while keeping 0.98 to 1.00 of values, then truncate below roughly twelve tokens per step; and ablating J-space [Gurnee et al., 2026]

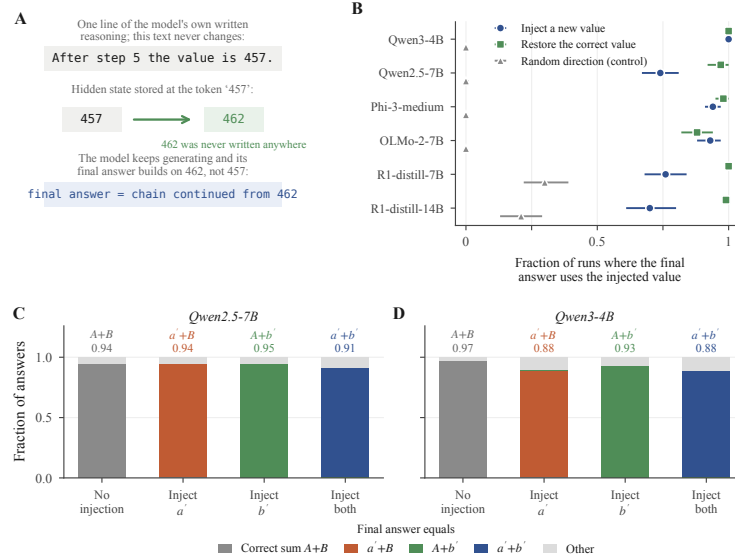


Figure 3: The hidden state under a written value is a settable register, and two registers compose. (A) The text still says 457; the hidden state at that token now encodes 462, which appears nowhere, and the final answer continues from 462. (B) Fraction of runs whose final answer uses the injected value; bars are 95 percent bootstrap intervals. The reasoning-distilled models re-solve after their reasoning phase, raising their random-control rate; re-derivation can only produce the correct answer. (C, D) With two counters A and B , injecting a replacement for either or both (a' , b') makes the answer equal the matching sum; answers using only one of two injected values occur at 0.01.

changes nothing at the strongest dose preserving ordinary generation (next-token agreement 0.87, perplexity ratio 1.12). The J-space ablation removes the top 16 active concepts at every position and leaves chain-of-thought accuracy, bare-answer accuracy, and the amount written unchanged, on synthetic chains and on GSM8K, doing no more than a size-matched random-directions ablation. The null holds at one to two steps, where models do perform the arithmetic internally.

This bounds the particular readout tested rather than the workspace as a whole: the dose is capped at generation-preserving strength, and a written token's contextual representation and its J-space component are not exclusive alternatives, since the lens samples the same residual states our patches overwrite. The workspace also exceeds J-space by construction: attention and MLP interiors and the KV cache fall outside the lens's domain, and within the residual stream the lens spans a fitted concept dictionary rather than the full state. Our register result is an existence proof of the gap: the planted value is causally decisive, sits in residual states the lens samples, and survives removal of the top 16 concepts.

4 Written tokens store causally active intermediate values

4.1 Restoring and replacing stored values

Figure 3 decomposes the effect of editing a written value. Restoring the correct hidden state while the text still shows a corrupted value returns the answer to correct in 0.88 to 1.00 of cases. Planting a third value that appears nowhere in the text redirects the answer to it in 0.70 to 1.00 of cases. The size-matched random control sits at or below 0.01 for the four instruction-tuned models. Because different planted values produce answers following those values, the value content is the causal quantity, and because the planted value is a mid-chain intermediate the model never produced, the patch does not inject the answer. The effect persists to 40-step chains (0.98 to 0.95 on Qwen3-4B). Rates are conditioned on items where the text edit was followed (0.44 to 0.98 by model; full table in the appendix).

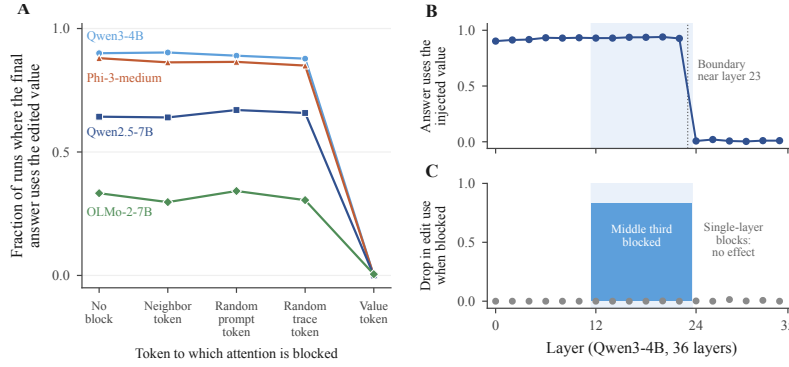


Figure 4: Written values are retrieved through attention to their token, across the middle third of layers. (A) Blocking later positions’ attention to the edited value’s token drops edit use to zero on four models; matched control blocks change nothing ($n=600$ per condition). (B) Overwriting the hidden state at any single layer up to 22 redirects the answer; from layer 24 on it does nothing (Qwen3-4B, 36 layers, $n=60$). (C) Blocking attention across layers 12 to 23 (shaded in both panels) removes 0.83 of edit use; no single layer’s block matters (gray dots, $n=80$).

On the narrative task the pattern replicates: a planted never-written count is followed at 0.83 [0.75, 0.90] on Qwen3-4B and 0.86 [0.79, 0.92] on Phi-3-medium, restoring the correct state returns the answer at 0.89 and 0.94, and the random control is 0.00 in both ($n = 86$ and 89).

4.2 Independent storage and composition

In a task with two counters summed at the end (Figure 3C, D), planting a never-written value at one final position yields the corresponding mixed sum (0.94 and 0.93 on Qwen2.5-7B, 0.89 and 0.93 on Qwen3-4B). Planting both yields their joint sum at 0.90 and 0.88, at or above the product of the single-position rates (0.89 and 0.83). The random patch leaves the clean answer in place (0.96 and 0.95), and answers using only one of the two planted values occur at 0.01. The reported number must be assembled at answer time from two values set independently, at different positions, and never written anywhere.

5 How models retrieve written state

5.1 Attention-blocking experiments

Blocking attention from all post-edit positions to the edited value’s position collapses the answer’s dependence on that value on every model tested (Figure 4A). Matched blocks of the adjacent words, of a random prompt position, or of a random position elsewhere in the trace leave the dependence at its unblocked level. Blocking the operand the next step needs breaks the arithmetic itself, yielding answers that match neither the edit nor the correct value. The collapse holds across three families and on both task formats (natural-language task: 0.730 to 0.007 on Qwen3-4B, 0.720 to 0.003 on Phi-3-medium).

Under the value-position block the unparseable-output rate stays at 0.000 (0.040 on OLMo-2-7B, equal to its unblocked rate). The model produces a fluent, parseable number matching neither value: it computes forward from state it can no longer retrieve. Reading is the preferred path rather than the only one: replacing the value’s characters with unreadable dots, leaving attention untouched, also removes following, and the Qwen models then reconstruct the value from the previous line in 0.60 to 0.73 of cases while Phi-3-medium never does (0.00).

5.2 Where retrieval occurs across layers

Layer-resolved versions of both interventions locate the read in depth on Qwen3-4B (36 layers; Figure 4B, C). Overwriting the value’s hidden state at any single layer from 0 through 22 redirects the answer at 0.90 to 0.94, while single-layer patches at layer 24 and beyond redirect at 0.02 or

less ($n = 60$): the register can be rewritten anywhere upstream of its consumers and nowhere downstream of them. Blocking the value-position attention only at the middle third of layers (12 through 23) collapses following from 0.857 to 0.025, blocking only the early or late third leaves it unchanged, and no single layer’s block matters ($n = 80$): the read is distributed redundantly across the middle third and confined to it. The two sweeps agree on the same boundary, near layer 23. Coarse sweeps on two further models (appendix) show the same structure with model-specific boundary depth.

6 When models retrieve versus recompute

Models read the register by default. On DeepSeek V3.2, an edited value the prompt merely implies is followed at 0.93, while an edited value the prompt generatively defines (the prompt states the two numbers whose sum becomes the running value) is reverted at 0.43; with the defining sentence present but the edit two steps away, following returns to 0.96, so the reversion targets the defined value. Claude Sonnet 4.5 recomputes more broadly, reverting most when a stated checkpoint is the value’s only source. Llama 3.3 70B follows edits at 0.97 whether or not the value is prompt-defined, despite solving the same chains from scratch, so the capacity to recompute does not imply recomputing. Qwen models at 4B and 7B never revert. The conditions come from separate experiments, so we claim within-experiment contrasts only.

The same test bounds the mechanism on natural benchmarks and bears on a disputed question. On GSM8K worked solutions, an edited intermediate is followed at 0.87 by Qwen3-4B and at 0.10 by V3.2, since a benchmark intermediate is often one operation from numbers still in the prompt and a stronger model recomputes it. Within one model, V3.2 follows edits at 0.93 on chains it cannot regenerate and at 0.10 on GSM8K. Lanham et al. [2023] report reliance on written reasoning declining with scale on standard benchmarks; our contrasts indicate the direction depends on the state itself. On regenerable state, stronger models depend less on the trace; on state no model can regenerate, dependence is high at every scale we test. Faithfulness of the trace to the computation is a property of the state’s regenerability.

7 Related work

Our interventions build on causal tracing and activation patching [Meng et al., 2022] and causal abstraction [Geiger et al., 2021]; Heimersheim and Nanda [2024] caution that broad patches conflate a component mattering with deleting information at a position, which the single-layer sweep in Section 5 addresses. Faithfulness studies [Turpin et al., 2023, Lanham et al., 2023] and filler-token results [Pfau et al., 2024] bound our claims to state the model does not regenerate internally.

8 Limitations

Tasks are numeric and count state tracking, so our claims concern serial task state rather than working memory in general; code, plans, and multi-entity state are untested. The J-space result bounds one fitted lens at generation-preserving doses and does not measure the J-space content of the written positions themselves. The fine layer sweeps are established on one model, and the coarse cross-model sweeps do not resolve Qwen2.5-7B’s read boundary. White-box evidence is at 4B to 14B; frontier evidence is behavioral. The only patch control is size-matched noise; a structured control planting a valid activation for a different quantity would further isolate value content.

9 Conclusion

Serial reasoning state outgrows a language model’s internal workspace within one to two dependent steps at every scale tested, and no intervention shifts state between the workspace and the trace, so the chain of thought is working memory rather than overflow. It works as compositional registers: settable by intervention, retrieved through attention in the middle layers, independently addressable, and decisive for the answer. For oversight, the written positions are a causally used memory channel, yet identical text can carry different stored values, so trace monitoring observes the channel without verifying the state.

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